PLAGIARISM / SIMILARITY REPORT

Manual Academic Integrity and Originality Assessment

Document Under Review

Report Title	Approximation Methodologies for the Travelling Salesman Problem: A Combinatorial Optimization Survey
Subtitle	A research-oriented survey with a validated experimental demonstration
Students	MARAPALLY RUUTHWIK (Enrollment No. 92400118870) ANNAVARAPU SRIVEK (Enrollment No. 92460118171)
Institution	Marwadi University Department of Computer Science and Engineering – Artificial Intelligence and Machine Learning (CSE – AI/ML)
Course	Design and Analysis of Algorithms (DAA) – Project Report
Project Guide	Mr. Sandeep Panchal
Academic Year	2026–2027
Document Length	40 pages (front matter, chapters, references and appendices)
Review Date	04 November 2026

1. Review Methodology

This assessment examines the submitted project report for textual similarity, possible unattributed reuse of published material, internal repetition, citation practices, and overall academic originality. The review consists of a full manual reading of the document (abstract, all chapters, tables, figure captions, references and appendices) together with targeted web searches for the exact report title, the names of the student authors, and distinctive experimental statements that appear in the Abstract and Chapter 7.

Important note on method: *No commercial plagiarism-detection database (Turnitin, iThenticate, Ouriginal, DrillBit, or institutional repositories) was accessible for this review. Consequently this document is a* ***manual similarity and originality assessment only****. No automated similarity percentage is reported, and no matched-source list from a commercial system is claimed.*

2. Web-Search Findings for Distinctive Content

Searches were performed for the exact report title, the student names MARAPALLY RUUTHWIK and ANNAVARAPU SRIVEK, and several highly specific experimental claims (including the statement that farthest insertion was the best constructive method, the reported mean excess figures of 34.6 % and 5.2 % at 200 cities, and the observation that proved bounds held on 300 metric test instances but were violated on random non-metric matrices).

None of these distinctive strings returned matching published papers, preprints, institutional repositories or other student reports. Standard textbook and survey literature on the Travelling Salesman Problem (double-tree algorithm, Christofides' algorithm, Lin–Kernighan, Held–Karp bound, etc.) appears widely online, as is expected for a classical survey topic. The present report frames such material as a literature survey and supplies citations.

3. Content Analysis

3.1 Nature of the Work

The document is a research-oriented undergraduate survey of approximation methodologies for the classical Travelling Salesman Problem, accompanied by a validated experimental demonstration implemented in pure Python. It classifies exact, approximation, constructive, local-search, metaheuristic and learning-assisted methods, presents the classical 2-approximation and 3/2-approximation results with proofs under the metric-symmetric assumptions, and reports original measurements on random Euclidean instances up to 200 cities and on 300 metric versus non-metric test instances.

3.2 Citation Practice

The report maintains a numbered reference list and cites foundational sources (Dantzig–Fulkerson–Johnson, Held–Karp, Christofides, Serdyukov, Rosenkrantz–Stearns–Lewis, Lin–Kernighan, Helsgaun, Johnson–McGeoch, Arora, Karlin–Klein–Oveis Gharan, Svensson–Tarnawski–Végh, and others). Classical algorithmic descriptions and proofs are presented in the authors' own wording and are accompanied by explicit statements of the assumptions under which the guarantees hold. No large blocks of unattributed verbatim text from a single identifiable source were observed during the manual reading.

3.3 Experimental Component

Chapter 7 describes a self-contained Python implementation with automated tour validation and independent cost recomputation. The concrete numerical results (mean gaps, run-time tables, the 8-city demonstration instance, the 300-instance metric/non-metric test, and the scaling plots) appear to be generated by the authors' own code. These specific figures and claims were not located in published literature under the same wording. The report repeatedly distinguishes proven approximation ratios from observed gaps.

3.4 Internal Consistency

The document exhibits normal academic repetition of key definitions and statements of assumptions across chapters (standard for a survey). No evidence of large identical blocks copied from one section to another was found. Tables and figure captions are consistent with the surrounding prose.

3.5 Observations on Standard Material

- Short descriptive passages about well-known algorithms (nearest-neighbour, 2-opt, MST doubling, Christofides' matching step) inevitably resemble textbook treatments. This is expected in a survey of a classical topic. The authors mitigate the issue by supplying citations, stating assumptions clearly, and adding their own experimental validation.
- Learning-based methods are reviewed from the literature only; no neural implementation is claimed. This limitation is stated explicitly in the Scope section.
- The pure-Python implementation is intentionally modest (instances up to 200 cities). The report does not claim to compete with state-of-the-art solvers such as Concorde or LKH; this is appropriately caveated.

4. Summary of Findings

Criterion	Observation	Assessment
Exact title / student names found online	No matching published or student documents located	No match found
Distinctive experimental claims	Specific percentages, 300-instance test and implementation details not located elsewhere	No match found
Classical algorithmic content	Standard survey material; presented with citations and assumptions stated	Expected for topic
Citation coverage of key results	Foundational and recent landmark papers are referenced	Satisfactory
Unattributed verbatim blocks	None identified in the manual review	None observed
Distinction between proof and experiment	Explicitly maintained throughout the report	Good practice
Overall character of the work	Survey of established methods plus original validated experiments; suitable for the stated DAA project requirement	Consistent with requirement

5. Conclusion of the Manual Assessment

On the basis of a full manual reading of the submitted report and targeted web searches for its distinctive content, no evidence of plagiarism or unattributed copying of substantial text was identified. The work is a properly framed undergraduate survey that combines standard combinatorial-optimisation material (with citations) and an original, validated experimental demonstration. The authors correctly separate proven approximation ratios from observed empirical gaps and state the assumptions required by each guarantee.

This assessment is limited to the manual methods described above. Institutions that require an official similarity score from a licensed commercial system (Turnitin, iThenticate, or equivalent) should obtain that score through their institutional account. The original project report PDF has not been altered, edited, paraphrased or replaced in any way by the production of this separate review document.

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